2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
October 1-5, 2023. Detroit, USA

Characteristics of Permanent Magnet Coupling based Wireless Manipulation via Simulation*

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***Abstract*— Characteristics of wireless manipulation based on permanent magnet coupling, including anchoring distance, panning torque, and translational force, are assessed in this paper. The study focuses on a typical scenario where a slave robot embedded with a small permanent magnet can be remotely controlled within a constrained area by a master large permanent magnet placed outside the environment at a certain distance from it. The key parameters (force and torque) acting on the slave robot are quantified and evaluated. In this article, several combinations of permanent magnets with various dimensions and configurations are studied using finite element methods. Based on the obtained results, we create a lookup table for each parameter, serving as a guideline to help interested researchers choose suitable magnetic combinations for their applications.**

I. INTRODUCTION

Wireless manipulation using remote energy, such as optical or magnetic fields, has attracted widespread attention in recent years [1]–[8], with targeted agents ranging from single cells to centimeter-scale micro-robots. In the biomedical community, utilizing magnetic coupling based remote control in disease diagnostics [9], drug release [10], and assisted surgery [11], [12], has been popular for decades [13] and has brought obvious benefits. Without direct contact, the desired force and torque could be generated by the interaction of magnetic fields between the master and slave agents. The magnetic field could be generated from various materials, among which permanent magnets and electromagnets are the two main approaches acting as master-driven sources. The former has larger energy density, power efficiency, and is capable of generating a stronger magnetic field with a smaller volume and mass [14].

The slave agents that respond to the motion induced by the master magnetic field, as shown in Figure 1, can vary and include permanent magnets [15]–[17], electromagnets [18], soft magnets [19]–[21], and ferromagnetic micro/nano-particles [22]–[25] depending on the specific tasks. The pure permanent magnet-composed slave agent is also expected to generate maximum force and torque in small dimensions

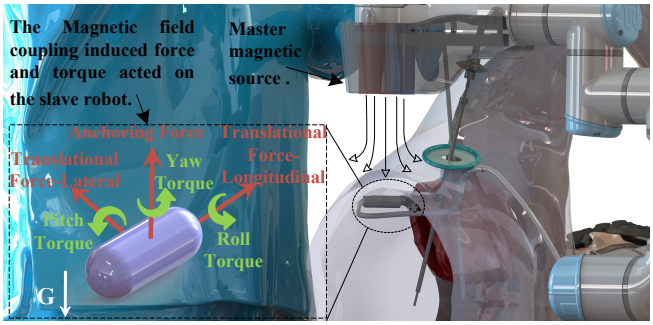


Fig. 1. The magnetic force and torque acted on the slave robot that is studied in this paper. The background of this scenario is the magnetic anchored and navigated intelligent endoscope system developed by this team for single-port video-assisted thoracic surgery (VATS).

[26]. In this paper, we focus on investigating the interaction between permanent magnets-based master and slave actuation systems to manipulate a centimeter-scale slave robot at a substantial range with sufficient force and torque in a confined environment. The geometric structure and magnetic field of the master and slave agent, which compose the "two-magnet" system, can be constructed and radiated by either a single or multiple assembled permanent magnet(s), depending on the specific application's requirements.

There have been various "two-magnet" magnetic manipulation systems published in recent years [27]–[31]. In [27], Yim and Sitti demonstrated a slave endoscope with embedded one pair of permanent magnets (two cylindrical NdFeB magnets with 1.0 T residual magnetic flux density, 10 mm diameter, and 6 mm thickness). This system can work inside the stomach and be remotely actuated by an external master permanent magnet (one cylindrical NdFeB magnet with 1.4 T residual magnetic flux density, 50 mm diameter, and 80 mm thickness). The controllable anchoring force (0.7∼1.5 N) and torque moment were the key characteristics to achieve the desired locomotion and drug release function. Meanwhile, the external permanent magnet (EPM) was manually controlled, and no other ferromagnetic instrument was operated near it. In [28], Liu et al. developed a magnetic actuated robotic camera system that utilizes three axially magnetized cylindrical permanent magnets (two N52 side magnets with a diameter of 25.4 mm and a thickness of 25.4 mm, and one central magnet with a diameter of 22.22 mm and a height of 28.57 mm). The system also includes two electromagnets, each with a height of 50 mm, an outer diameter (OD) of 50 mm, an inner diameter (ID) of 10 mm, an iron core with a height of 60 mm, AWG23

*This work is supported by Hong Kong RGC General Research Fund with project number 14203019, 14202820, and 14214322, multi-scale medical robotics center, and partially supported by International Science and Technology Cooperation Project of Guangzhou Economic and Technological Development District (Grant No. 2020GH01).

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2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) | 978-1-6654-9190-7/23/$31.00 ©2023 IEEE | DOI: 10.1109/IROS55552.2023.10341587

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copper wire with 2000 turns, and an input current of up to 1.5 A, which act as external actuators. The internal slave follower consists of three diametrically magnetized cylindrical permanent magnets (two wing-shaped tail magnets with an OD of 12.7 mm, an inner diameter of 4.75 mm, and a thickness of 6.35 mm; and one central magnet with a diameter of 6.35 mm and a length of 12.7 mm). The main parameters that determine the motion of the camera system include the anchoring force (1∼4 N), lateral translation force (∼0.1 N), panning torque (∼0.01 N·m), and tilt torque (up to ∼2.5 mN·m). The hybrid integrated external magnetic system enables the decoupled control of the tilt and translation motion. In [29], [32], the authors developed a single port laparoscopic camera based on magnetic linkage. The former proved a concept with a magnetic link that employed one cubic magnet (for anchoring) and one diametrically magnetized cylindrical magnet (for torque) as EPM (30x30x30 mm, magnetized along the Z-axis, N42; 9.5 mm radius, 19 mm length, N42), and one diametrically magnetized cylindrical magnet as IPM (6 mm radius, 13 mm length, N52). The latter presented an orthogonal magnet arrangement-based laparoscopic camera with one permanent magnet (diametrically magnetized ring-shaped with 25.4 mm OD, 6.35 mm ID, 25.4 mm thickness, and 1.32 T strength) as an external driven source and two permanent magnets (one spherical magnet with 4.1 mm diameter and 1.32 T, one diametrically magnetized magnet with 3.155 mm diameter, 6.35 mm thickness, and 1.32 T strength) as internal correspondents. With up to 3.82 N anchoring force, 2.95 N of translation force, pan, tilt, and auto-flipping motions were achieved to control the endoscope. In [30], Norton et. al demonstrated a magnetic servoing of a capsule robot for gastrointestinal ultrasound. The manipulation system was composed of one external EPM and one internal IPM, both cylindrical, made of NdFeB, and axially magnetized at N52 grade. The external magnet was 101.6 mm in diameter and length, and the internal magnet permanent magnet was 11.11 mm in diameter and length. The magnetic forces were properly adjusted (0.25∼2.5 N, then constant at 0.6 N) to achieve the desired robotic capsule endoscope's tilt orientation ($\pm 10^{\circ}$). In [31], Xu et.al presented a magnetic actuated flexible-joint laparoscope with three permanent magnets (two cubic magnets with 20x20x20 mm, one ring-shaped, diametrically magnetized magnet with 28 mm OD, 12 mm ID, and 30 mm thickness) as an external actuator and three permanent magnets (two cubic magnets with 10x10x10 mm, one cylindrical, diametrically magnetized magnet with 10 mm diameter, and 8 mm thickness) as internal correspondents, where all magnets were made of NdFeB and magnetized at N52 Grade. With this combination, the endoscope can be anchored at a separation distance of 40 mm with a tilt angle up to 60°.

In addition to the above-reviewed work, our previous investigation [33] also revealed a feasible magnetic combination. Here, we have listed the main designs ever employed in Table I to better understand the application features. The abbreviations Ref., Qua., Mag., App., Axi., Dia., Thi., and Sph. represent Reference Number, Employed Quantity, Direction of Magnetization, Application, Axial, Diametral, Thickness, and Spherical Magnetization, respectively. It can be seen that typical application scenarios include stomach/gastrointestinal disease diagnosis and treatment, as well as endoscopic minimally invasive surgery (laparoscopic/thoracic).

TABLE I

COMPARISON OF TYPICAL "TWO-MAGNET" MAGNETIC ENDOSCOPE MANIPULATION SYSTEMS FROM PUBLISHED LITERATURE.

Ref.	Master Magnet(s)		Slave Magnet(s)		App. Scenario
	Qua.	Mag.	Qua.	Mag.	
[27]	1	Axi.	2	Axi.	Stomach Disease Diagnose and Treatment
[28]	5	Axi.	3	Dia.	Single-Incision Laparoscopic Surgery
[29]	2	Thi., Dia.	1	Dia.	Laparoendoscopic Single-Site Surgery
[32]	1	Dia.	2	Sph., Dia.	Minimally Invasive Surgery
[30]	1	Axi.	1	Axi.	Gastrointestinal Ultrasound Endoscopy
[31]	3	Thi., Dia.	3	Thi., Dia.	Single-Incision Laparoscopic Surgery
[33]	3	Dia., Axi.	3	Dia., Axi.	Video-Assisted Thoracic Surgery

The above-published literature presents and validates several possible configurations of the "two-magnet" manipulation system, with the number ranging from single to multiple assembled magnets. These configurations can function as either actuator magnets or correspondent magnets. Besides the quantity factor, the shape and associated direction of magnetization of the magnets will determine the degree of freedom (DoF) that the system is capable of. Meanwhile, the internal space between the component magnets shall affect the magnitude of the coupled magnetic forces and torques. Lastly, it is worth indicating that many other geometries, dimensions, and combinations could be used and would be reasonable, but they are not covered by this section's review.

The structure of this paper is arranged as follows. This section introduces the background of this work. Section II lists the possible shapes and magnetic field directions for master and slave magnet(s). Geometric constraints that need to be considered are also included. Section III introduces the methods employed in this paper. Section IV demonstrates the results obtained. Possible improvement suggestions for application scenarios that appeared in the literature are also proposed in this section. Section V concludes this paper and envisions future work.

II. DIFFERENT CONFIGURATIONS OF MAGNETIC ACTUATION SYSTEM

In this section, we have listed twelve permanent magnet configurations that have already been applied or could be

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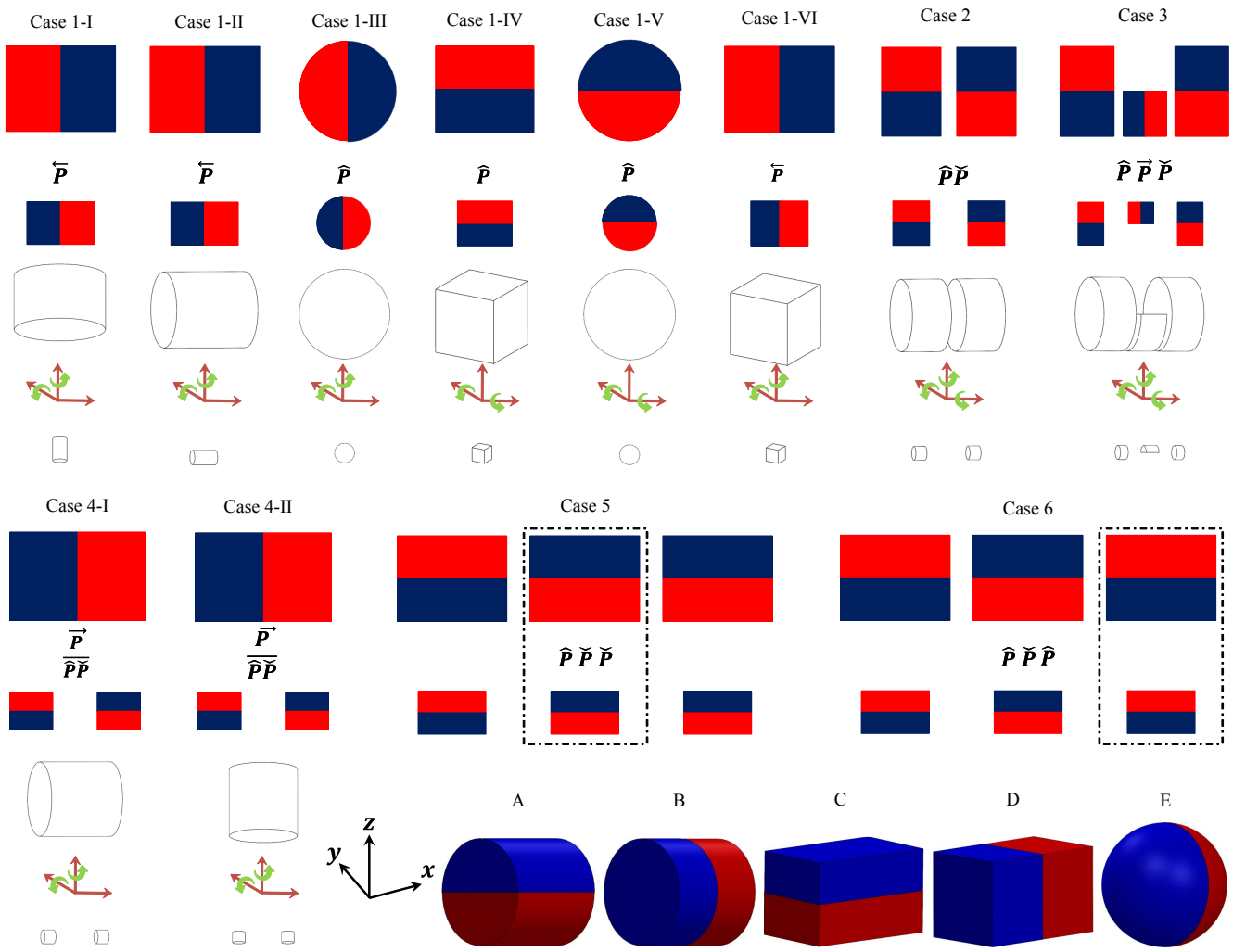


Fig. 2. The "two-magnet" scenarios that studied in this paper.

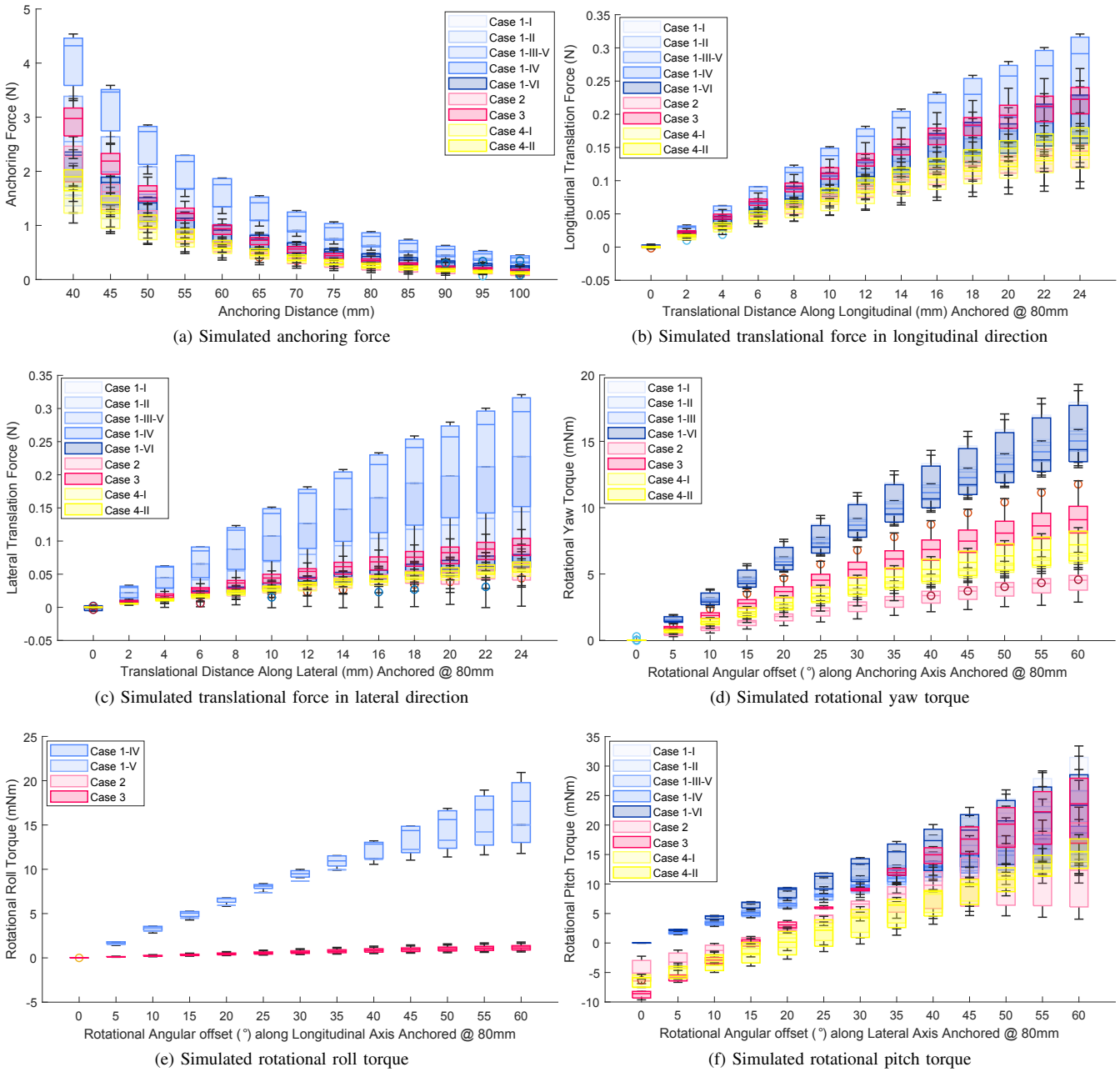
employed as a "two-magnet" magnetic actuation system to fulfill certain tasks. The geometric and mass constraints that guarantee an impartial comparison are also stated in this section.

A. Master-Slave Permanent Magnet Configurations

Based on the demonstrated capabilities and specific application scenarios' demands in published literature, we have listed twelve configurations that are arranged as master and slave robots, as shown in Figure 2. The magnetization direction, employed geometric shape, and various arrangements are indicated and presented from Case 1-I to Case 6. The character representation of each system arrangement, from $\overleftarrow{P}$ of Case 1-I to $\hat{P}\check{P}\hat{P}$ of Case 6, is denoted. For intuitive understanding purposes, the specific shapes studied in this article, along with their corresponding magnetization direction, are also illustrated and can be referred to as A to E. Meanwhile, a simple coordinate system indicator is also exhibited. The detailed definition of the coordinate system can be referred to in our previous work [33].

1) Single-to-Single Configurations: The most compact configuration of the "two-magnet" actuation system is single-to-single, as shown from Case 1-I to Case 1-VI. In Case 1-I, the employed permanent magnets are two cylinder-shaped with their main axes aligned and diametrically magnetized (type A). This type of configuration is capable of five degrees of freedom (DoF) without roll rotation. In Case 1-II, the employed permanent magnets are two cylinder-shaped with their main axes parallel and axially magnetized (type B). This type of configuration is also capable of five DoF. In Case 1-III, the employed permanent magnets are two spherical-shaped with their magnetic fields cross-mapped (type E). This type of configuration is also capable of five DoF. In Case 1-IV, the employed permanent magnets are two rectangular cuboid-shaped with their magnetic fields directly mapped and arranged (type C). This type of configuration is also capable of five DoF but loses the yaw rotation. For Case 1-V, the employed permanent magnets are two spherical-shaped with their magnetic fields directly mapped and arranged (type E). Similar to Case 1-IV, this type of configuration also loses the DoF of yaw rotation. The employed permanent magnets

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(a) Simulated anchoring force

(b) Simulated translational force in longitudinal direction

(c) Simulated translational force in lateral direction

(d) Simulated rotational yaw torque

(e) Simulated rotational roll torque

(f) Simulated rotational pitch torque

Fig. 3. Simulation results of the selected "two-magnet" scenarios studied in this paper are presented.

are two rectangular cuboid-shaped with their magnetic fields cross-mapped and arranged (type D) in the Case 1-VI scenario. For DoF, this type of configuration only loses the actuation capability of roll motion.

2) Multiple Magnets Assembled Configurations: From Case 2 to Case 5, the slave robot is composed of more than one magnet, a common design in many magnetic actuated capsule robots. In Case 2, both the master and slave magnet arrays consist of two diametrically magnetized cylindrical permanent magnets (type A) with their magnetic fields cross-mapped and arranged. This configuration, equivalent to combining two pairs of Case 1-II, provides six DoF. Case 2 also integrates a pair of half-cylinder-shaped axially magnetized permanent magnets (type B) with their magnetic fields repelled to the side magnets. This configuration, referred to as Case 3, has been studied and compared with other cases in our previous work [33], and it also offers six DoF. Case 4-I and 4-II are variants of Case 1-II and 1-I, respectively, obtained by equally splitting the slave magnet into two. These configurations still lack the capability of roll motion. Building upon Case 2, integrating one pair of Case 1-IV into the middle or side results in Case 5 and Case 6, which have already been applied in [34] and [31], [35]. These two configurations are also capable of full DoF, with the rotational motions achieved by the partial components' motion.

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TABLE II

SIMULATED RESULTS OF THE MAGNETIC FORCE AND TORQUE ACTED ON THE SLAVE ROBOT.

Case No.	Subcase No.	Magnetic Force (N)			Magnetic Torque (mN·m)			Performance Evaluation		
		Anchoring	Longitudinal	Lateral	Yaw	Roll	Pitch	DoF	Missing Motion	Comments
1	Case 1-I	0.9281	0.0913	0.0316	7.6	N/A	8.8	5	Roll	General use
		0.7834	0.0774	0.0272	6.4		8			
		0.5992	0.0595	0.0207	5.1		7.4			
	Case 1-II	0.7239	0.0689	0.0244	6.3	N/A	7.7	5	Roll	General use
		0.6600	0.0620	0.0218	5.9		6.6			
		0.4980	0.0474	0.0159	4.7		5.8			
	Case 1-III	0.6989	0.0663	0.0219	5.9	N/A	7.4	5	Roll	General use
	Case 1-IV	1.8761	0.1234	0.1234		6.8	6.8	5	Yaw	1st choice for anchoring, translation
		1.6183	0.1042	0.1085	N/A	6.5	6.5			
		1.2176	0.0751	0.0751		5.8	5.8			
	Case 1-V	1.3473	0.0878	0.0878	N/A	5.9	5.9	5	Yaw	2nd choice for anchoring, translation
	Case 1-VI	1.0213	0.0981	0.0317	7.4	N/A	9.4	5	Roll	1st choice for pitch control
		0.8482	0.0771	0.0265	6.1		8.6			
		0.6195	0.0607	0.0206	5.2		7.2			
2 and 4	Case 2	0.7849	0.0723	0.0331	2.4	0.7067	2.8	6	N/A	2nd choice for 6 DoF
		0.6041	0.0574	0.0226	1.8	0.5009	2.4			
		0.3598	0.0392	0.0117	1.1	0.3083	2.1			
	Case 4-I	0.5936	0.0605	0.0225	2.7	N/A	1.7	5	Roll	Not suitable for pitch control
		0.5232	0.0508	0.0197	2.4		0.4224			
		0.3991	0.0387	0.0126	1.8		-1.3			
	Case 4-II	0.7558	0.0746	0.0271	3.4	N/A	2.6	5	Roll	Not suitable for pitch control
		0.6479	0.0641	0.0230	2.8		0.00506			
		0.5049	0.0485	0.0182	2.1		-2.7			
3	Case 3	1.1187	0.1093	0.0449	4.8	0.6501	3.8	6	N/A	1st choice for 6 DoF
		0.9157	0.0894	0.0350	3.6	0.4433	3.1			
		0.6445	0.0660	0.0224	2.2	0.2550	2.4			

B. Geometric and Mass Constraints

Intelligent robotic manipulation of the master magnetic source is expected to become commonplace in the predictable future. However, in the early stages of biomedical and surgical applications, the locomotion of the magnetic actuation source was mainly controlled manually as a standard practice [27]. Therefore, we must consider this trend and constrain the mass of the master magnet, as the payload capacity of a robotic manipulator is limited.

To ensure a fair comparison of configurations with a unified standard, we set a restriction of 1000 grams for the external master magnet and 10 grams for the internal slave magnet(s). Additionally, considering potential confined working environments, we impose restrictions on the dimensions, especially the length of the internal capsule robot, which should not exceed 50 mm.

III. METHOD

In order to assess the locomotion performance of the above-mentioned "two-magnet" actuation system, we utilize the finite element method (FEM) to simulate the magnetic force and torque acting on the slave agent induced by the master magnetic field source's locomotion.

The anchoring distance is defined as the distance from the top surface of the slave agent to the bottom surface of the master source. The translation distance longitudinally is defined as the offset of the geometric center between the master and slave in this direction. The translation distance laterally is defined as the offset of the geometric center between the master and slave in this direction. The rotational yaw angle is defined as the angular offset between the master and slave about their axis in the anchoring direction. The rotational roll angle is defined as the angular offset between the master and slave about their axis in the longitudinal direction. The rotational pitch angle is defined as the angular offset between the master and slave about their axis in the lateral direction.

As Case 5 and 6 are just extensions of Case 2, we predict that the characteristics of these two types shall have similar performance to Case 2. Meanwhile, their real application introduces flexible joints, which lead to a longer dimension in length for the slave robot. Considering the length constraint of the internal capsule robot that was introduced in Section II-B, its force and torque characteristics are not simulated in this paper.

For a spherical shape, there is only one parameter (the radius) to control the volume of the object, which means that for the same material, once its mass is determined, the shape of the spherical object is unique. However, for other geometric bodies with their shape determined by more than one parameter, such as cylinders and rectangular cuboids, their shapes could vary. Therefore, for each case from Case 1-I to 4-II, besides Case 1-III and 1-V, we run at least five instances with variations in the master and slave magnet's dimensions (while keeping the volume unchanged to maintain the total mass at 1000 grams and 10 grams,

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respectively) as well as the spacing among magnet units (for slave magnets, keeping the total length no longer than 50 mm). Each studied instance contains around 4 million mesh elements to strike a balance between efficiency and accuracy. The magnetization strength of all magnets is set to N52 grade.

The correctness of the applied simulation method in this paper has been validated in our previous work [33].

IV. RESULTS AND DISCUSSION

We obtained the simulated results, as shown in Figure 3, in boxchart view. A lookup table that summarizes the main values of each case for each parameter as a guideline, allowing interested researchers to choose suitable magnetic combinations for their applications, is created (Table II). The dataset for each group, which contains three data in the table, represents the maximum, mean, and minimum values for each object at specific conditions (anchoring distance, offset distance, or rotated angle), respectively. The anchoring force presented in this table is the value obtained when the anchoring distance reaches 60 mm. The data presented in the columns of Longitudinal and Lateral are the values obtained when the translational offset reaches 8 mm, respectively. The data presented in the columns of Yaw, Roll, and Pitch are the values obtained when the corresponding angular offsets reach $20°$, respectively.

From Figure 3 and Table II, we can conclude that for the induced magnetic anchoring force, it is obvious that Case 1-IV could provide the largest force among all cases. The advantage of this configuration is also demonstrated in the longitudinal, lateral, pitch, and roll motion. Meanwhile, for the generated anchoring force, longitudinal translational force, lateral translational force, yaw torque, pitch torque, and roll torque, it can be summarized that the single-master-to-single-slave arrangement could generate significantly larger values compared to multiple magnet-based configurations, except for the cases that lost certain DoF.

The radar chart in Figure 4 also illustrates the strength of each case in an intuitive approach, allowing researchers to compare the strengths and weaknesses of each magnetic configuration. Considering the employed permanent magnets that were reported in [16], [30], based on the demonstrated results above, we would suggest replacing the external permanent magnet from type 1-II to 1-VI (i.e., replacing the cylindrical-shaped magnet with a rectangular cuboid or cubic-shaped magnet) with the same mass. This change should improve the slave robot's locomotion characteristics in terms of anchoring force, longitudinal force, lateral force, and pitch torque.

V. CONCLUSION AND FUTURE WORK

In this paper, we present a characteristic comparison of the magnetic force and torque induced by the master and slave permanent magnet's magnetic field coupling. We list several commonly applied magnet configurations and study their forces and torques via FEM-based simulation. The obtained results, presented in box charts, tables, and radar

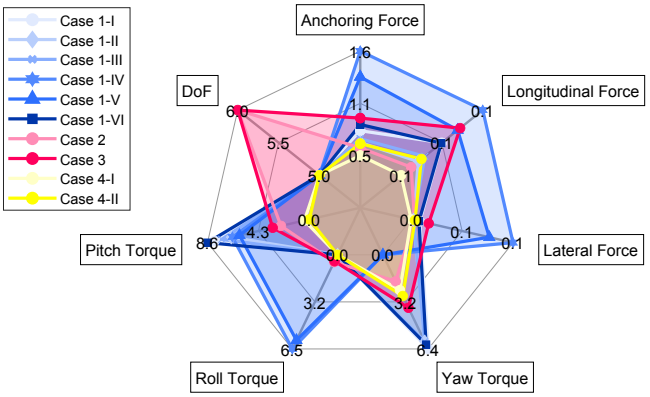


Fig. 4. The magnetic force, torque, and DoF are presented in the radar chart. The illustrated force and torque are the mean values listed in Table II.

charts, provide a useful map for researchers or engineers to select the proper configuration for specific applications.

For practical applications, based on the obtained results, we recommend applying Case 1-IV as the 1st choice and Case 1-V as the 2nd choice in translational (which depends on anchoring, longitudinal, and lateral force) demanding scenarios, and if yaw motion is not a compulsory DoF. For pitch-demanding exercises, Case 1-VI shall be the best choice. Meanwhile, for 6 DoF applications, we still suggest Case 3, but users should consider the assembly difficulty for this configuration as its internal repellent force is quite strong.

From the listed application scenarios in Table I, we can predict that magnetic coupling-based wireless manipulation will continue to explore its role in minimally invasive surgery, while the surrounding equipment (such as ferromagnetic instruments) might interact with the permanent magnet-embedded robot agent, whether it's the master or slave robot. This interaction could affect the real performance of the magnetic actuation system and could even prevent the system from working in confined environmental spaces. For future work, we will investigate what kind of magnetic arrangement could minimize the undesired magnetic interference with ferromagnetic instruments without sacrificing essential characteristics such as anchoring and locomotion capabilities.

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